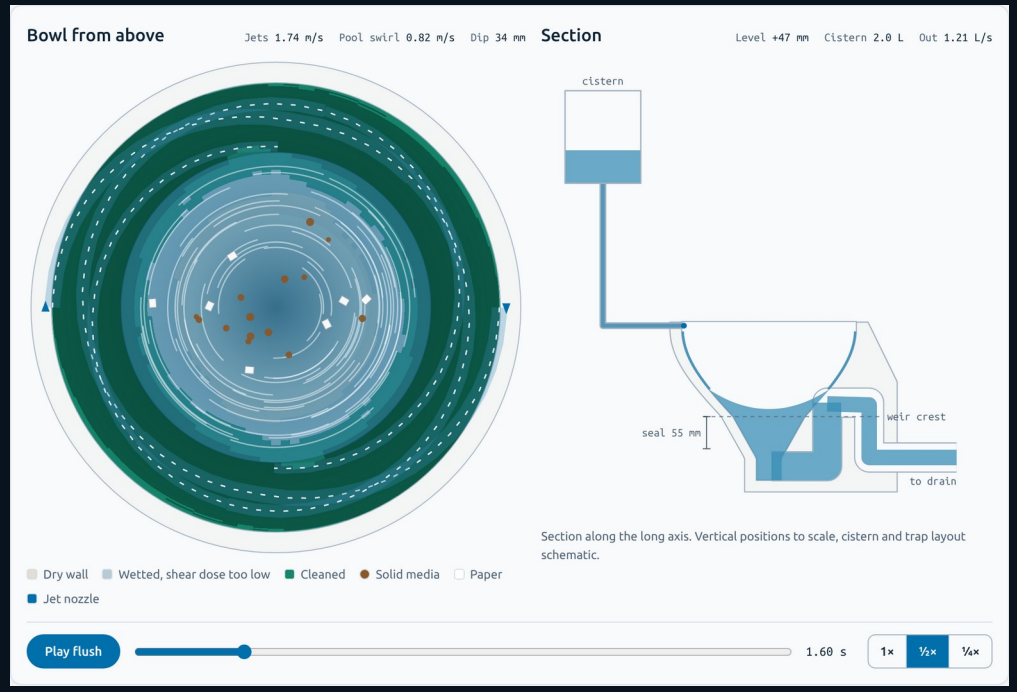
Faster design loops with a fast physics model and AI

A rimless toilet flush as a worked example: answer most design questions in seconds, and spend expensive simulation only where it decides.

6 fewer expensive iterations

per design question in the typical project of our challenge: 10.5 → 4.5, about 6 working days (synthetic reference, see slide 6)

Try it live fintlp.github.io/jet-flush-model



The fast model on the page: one flush, computed in 20 to 40 ms. Films on the wall, pool swirl, waste.

A demonstrator: the model is physics-based but not yet calibrated against real bowls; the challenge uses a synthetic reference; the CFD is a second simulation of one design. Each slide says what is measured and what is assumed.

THE PROBLEM

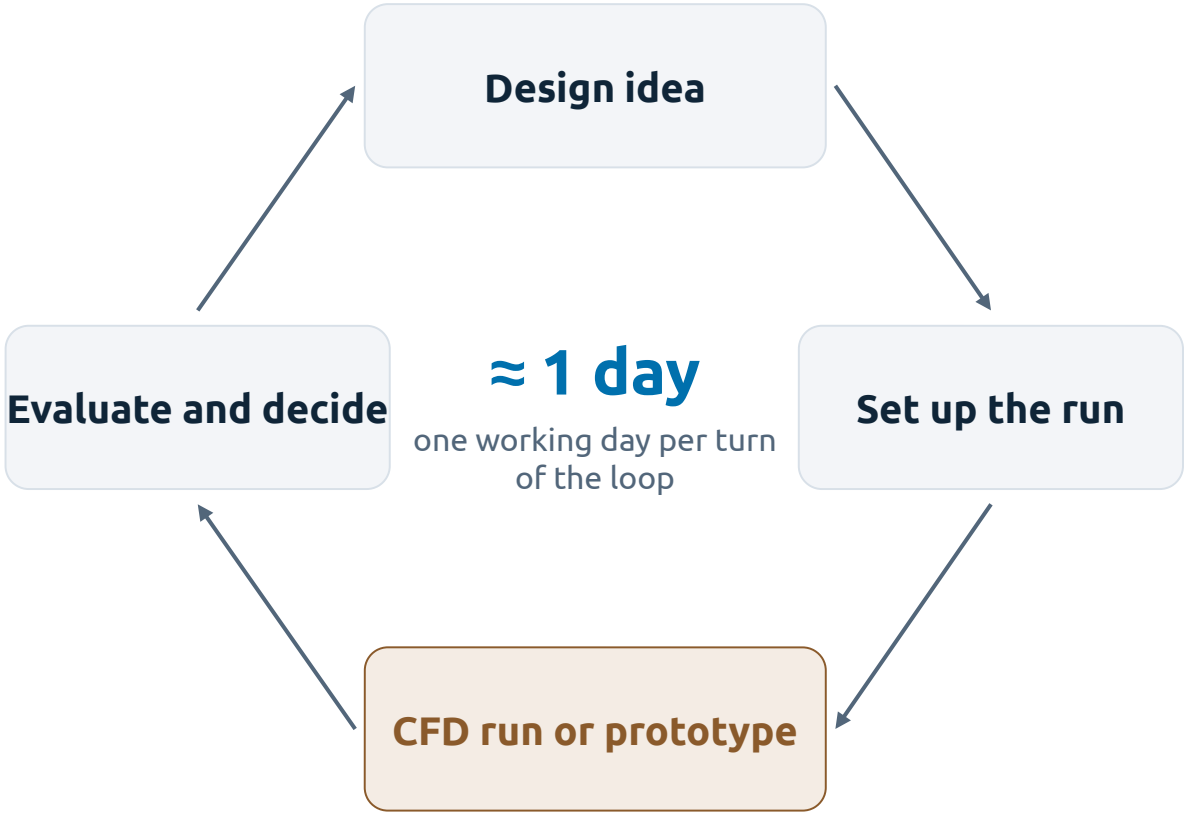
Design loops wait for expensive answers

Every design question waits. Each candidate goes through a high-fidelity simulation or a prototype: set it up, run it, evaluate it, decide.

About one working day per iteration. Geberit reported this for its CFD-led bowl development, with more than 100 virtual drafts before the first prototype.

So teams ask few questions. Many design decisions are still made on experience, before any number exists.

Late answers are expensive. A weak decision found late costs most to change: in the tool, the mould, the production line.



Source: CADFEM Journal, "Simulation leads Geberit to design a toilet bowl with asymmetrical geometry". The same pattern holds wherever design decisions wait for CFD, crash, rig or prototype tests.

THE APPROACH

Fast physics explores, AI narrows, high fidelity decides

1 **Fast physics model**
thousands of variants, milliseconds each

2 **AI on top**
optimise, learn surrogates, correct the model: tens of candidates

3 **High fidelity: CFD, rig, prototype**
a few finalists, hours to days each

every expensive run also calibrates the fast model

On the toilet flush

20–40 ms
per flush, fast model, in a browser

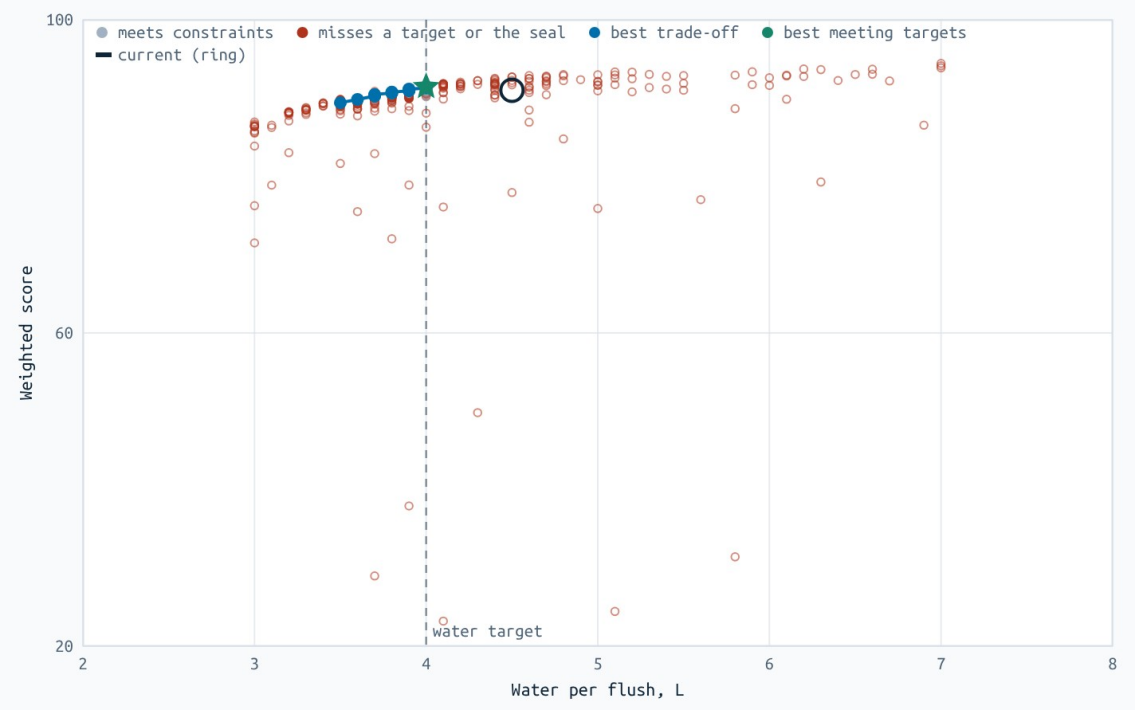
≈ 540
designs searched by the AI in about 10 s

≈ 2 hours
for 3.2 s of the same flush in OpenFOAM CFD, 12 cores

The answer is not to replace CFD or tests. It is to put the right tool at each stage and connect them, so that the slow tools are spent only on the designs that deserve them.

Design changes are tested while the meeting is still on

Change the bowl or the jets and see cleaning, waste removal, water use and noise at once. The AI lab searches for designs that meet set targets.



AI lab: each dot is one simulated flush; the curve is the best trade-off between score and water.

From the guided demo, part 1

1. **86 % → 79 %**
   wall cleaned when the bowl turns oval, jets at the ends of the long axis

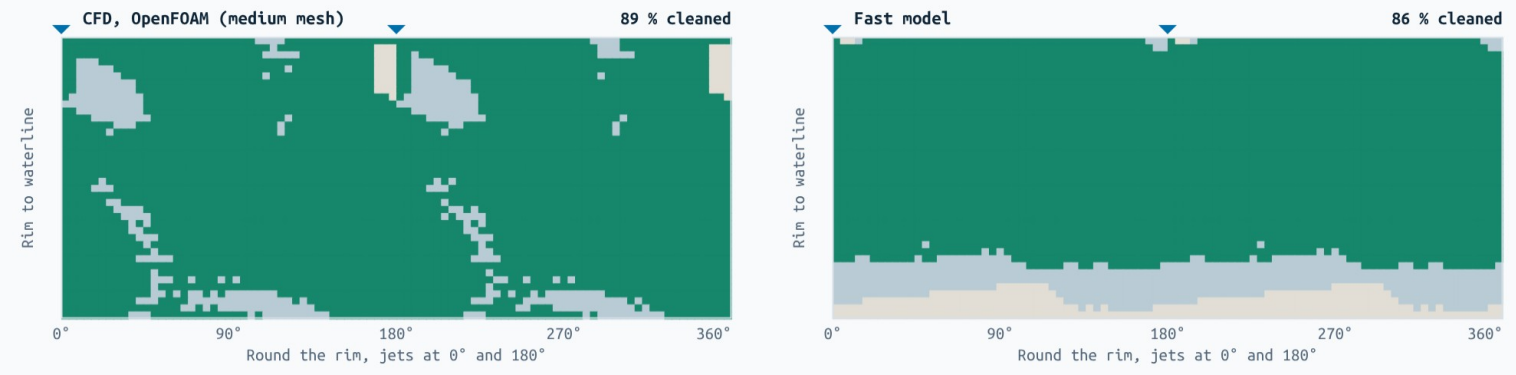
2. **→ 88 %**
   after moving the jets to the long sides

3. **89 % at 4.0 L**
   design found by the AI lab: 0.5 L less water than the start, −1.1 dB, no louder

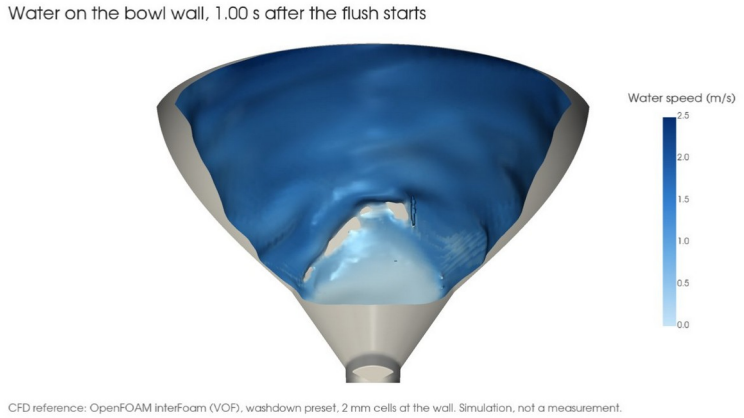
Illustrative estimates from the uncalibrated model, for comparing designs. Noise is a relative, experimental indicator. AI suggestions are candidates for a CFD or rig check, not final answers.

Against full CFD: close on the totals, different in the details

The same bowl and jets in OpenFOAM: about 2 hours on 12 cores for 3.2 s of flush, against about 30 ms for the fast model.



What gets cleaned, wall unrolled: CFD left, fast model right (green cleaned, grey wetted only).



CFD water surface at 1.0 s, front half removed.

89 % vs 86 %

of the wall cleaned, CFD vs fast model: close totals

0.14 s vs 0.54 s

first water in the pool: in the model it arrives later

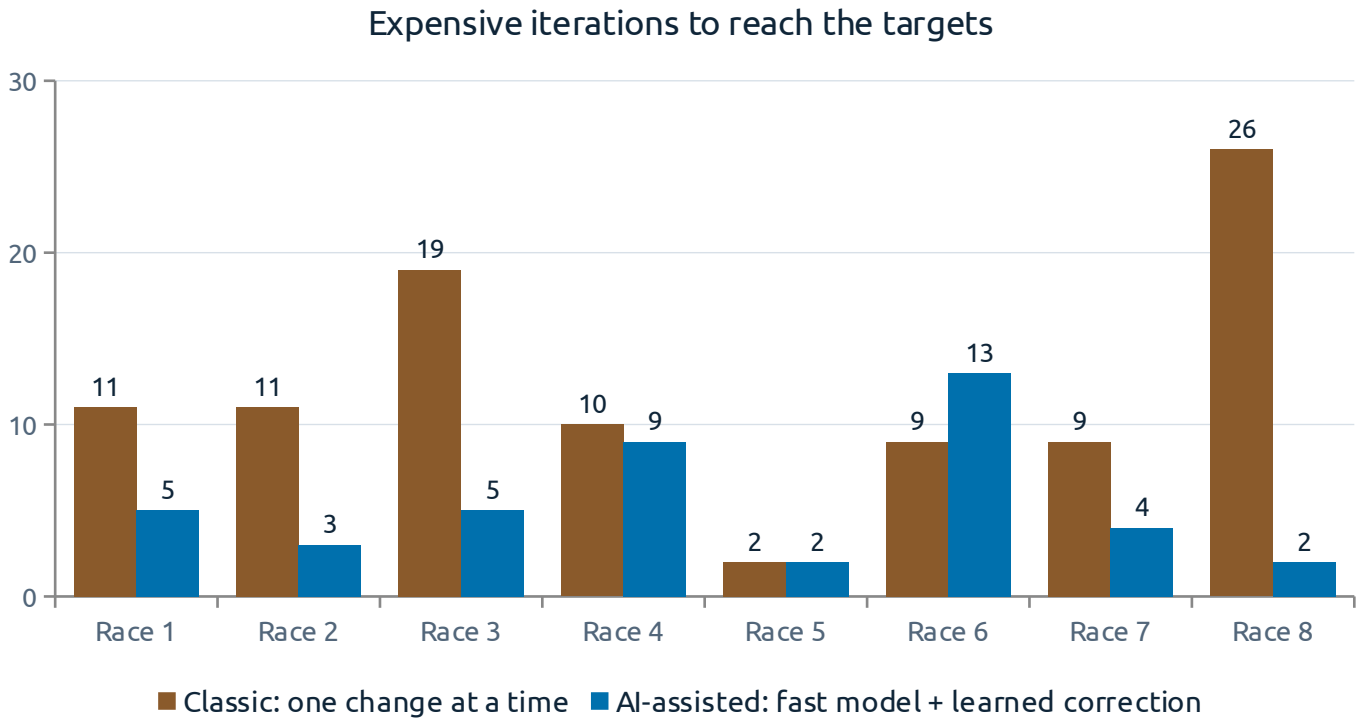
11 %

overlap of the uncleaned areas: the model gets the total, not where the gaps are

A second simulation of one design, not a measurement or a validation. Trap outflow taken from the model; after 1.3 s the CFD outlet draws air. On two CFD meshes the totals agree and the uncleaned areas overlap 50 %: a sensitivity, not mesh independence.

Learning from each run: less than half the iterations

The same design task, worked eight times: an oval bowl must reach 4.0 L and 87 % cleaning. Every candidate is checked with an expensive run.



10.5 → 4.5

iterations in the typical project (median of the eight races), classic against AI-assisted

6 of 8

races with fewer iterations for the AI route; 1 tie, 1 with 4 more

3 of 8

races the fast model alone never finished: without learning from the expensive runs it gets stuck

The expensive runs come from a synthetic reference (the fast model with other coefficients and effects it does not have), not from real CFD. The task has five settings; the classic route is a simple automated search, and design of experiments would need fewer runs.

WHAT IT IS WORTH

About 6 working days saved on every design question

The calculation, step by step, with example values. The calculator on the page takes your own figures.

10.5 iterations, classic (typical project) − **4.5** iterations, AI-assisted = **6** iterations fewer per design question × **1 day** per iteration (example: Geberit) = **6 days** working days saved per design question

8 h × 95 € person hours per iteration × cost per hour (examples) → **4,560 €** per design question: 6 × 760 € × **10 a year** design questions like this (example only) = **45,600 € a year** the break-even budget for building and calibrating a fast model

Iteration counts from the challenge (synthetic reference, eight races; race by race the saving ranged from −4 to 24, median 5.5, average 6.75). Examples: 1 working day per iteration from the Geberit report (CADFEM); 95 € per hour, average rate of freelance engineers in Germany (freelancermap, Feb 2026); 10 questions a year is an example only. Time to market is not priced in.

BEYOND THE TOILET

The toilet is the example, not the limit

The same pattern fits wherever design loops wait for simulation or tests. Read the calculator's words for your product:

Product area	Fast model	Expensive run
Sanitary ware (this demo)	Reduced-order flush model: jets, wall film, pool, trap	Free-surface CFD, or a ceramic prototype on the rig
Vehicle water management	Film and droplet paths over body, seals and drains	Full-vehicle CFD of rain or wading; climatic wind tunnel
Lubrication and cooling	Network model of oil flow and splash in gearbox or e-motor	Particle (SPH) simulation of oil splash; rig with a transparent housing
Appliances	Spray and drainage model of dishwasher or washing machine	CFD of spray arms or drum flow; wash test with a soiled load
Sensors and ADAS	Droplets and airflow over camera and lidar covers	CFD of droplets in airflow; wind tunnel spray test

One iteration is one variant through the expensive run and its evaluation. These rows are examples of where the pattern fits, not results: none of these fast models is part of the demo.

HONEST ABOUT THE LIMITS

What the demo shows, and what it does not show yet

Shown

The model solves its own equations correctly: 61 physics checks, from energy and water balance to grid and time-step convergence.

One check against full CFD: the same bowl in OpenFOAM on two meshes, with where the two differ.

The AI workflow end to end: search, surrogates, calibration and learning from expensive runs, all live in the browser.

Every figure is traceable: to a check script or to data stored with the page.

Not shown yet

Calibration against real bowls: the coefficients are physics-based but not fitted to rig data.

Design trends against CFD: one design was checked; trends need CFD runs of several designs.

Real CFD in the challenge: the expensive runs come from a synthetic reference.

Noise against recordings: noise is a relative indicator, fitted so far only to synthetic recordings.

Use the demo to compare designs and to decide what to test, not as validated values. The presets are generic EU washdown and siphonic bowls, not any manufacturer's product.

HOW A TEAM WOULD START

Start with one design question, in four steps

1

Pick one question

one recurring design question that today waits for CFD or tests

2

Build a fast model

of its main mechanisms, with the people who know them

3

Calibrate it

with CFD or test data you already have, and check it on cases it has not seen

4

Close the loop

optimisation on top, and every new expensive run correcting the model

In a workshop we take one of your design questions through these steps on paper: where the loop waits, which mechanisms a fast model needs, which data you already have to calibrate it, and what it would save with your own figures.

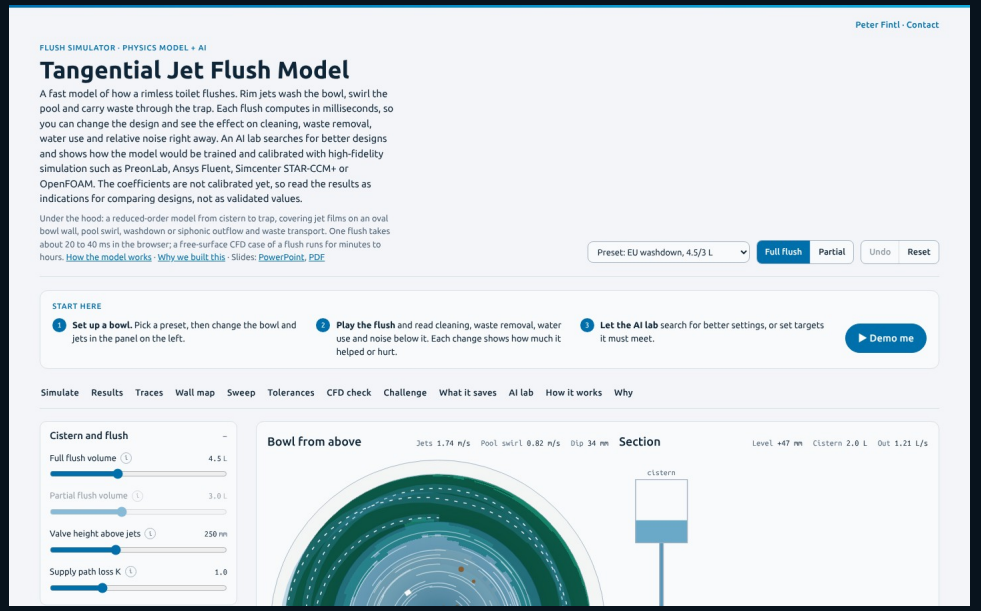
NEXT STEP

Try it, then bring us one of your design questions

1 **Try the page**
The guided story in four parts, the calculator and every tool, live in the browser. fintlp.github.io/jet-flush-model

2 **Watch the story**
All four parts, recorded at reading pace. The video is in the PowerPoint version of this deck.

3 **Contact Peter Fintl**
Ask for a workshop on one of your own design questions. fintlp.github.io/latest-news/#contact



The guided story, recorded from the page (v2.1.0) on 22 October 2026.

Scan to open the live page

A demonstrator with a physics model that is not yet calibrated; synthetic references are labelled on the page; the CFD is a second simulation of one design, not a measurement.